

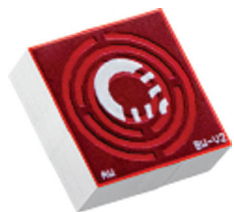
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Antenna designed to enhance coverage in all directions

The BodyWave antenna is an on-ground SMD antenna optimised for wearable applications. The BodyWave increases radiation efficiency, detuning immunity and propagation in all directions. Body-



Wave antennas are designed to increase coverage in directions obscured by the human body, with 10 to 20dB increased

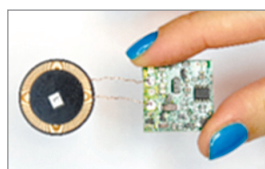
link budget in those directions when compared to existing commercially available chip and printed antennas. It features a 50-ohm SMD antenna component. This device can allow designers to create wearable devices that work without connectivity issues even in crowded rooms. The BodyWave antenna can produce BodyWaves that help facilitate wireless communications when there is no direct line of sight with both on-body and off-body receivers. It is designed for use in the 2.45GHz ISM frequency band, suitable for Bluetooth, Wi-Fi, and ANT+ applications.

Antennaware
<https://antennaware.co.uk>

Innovative wireless power solution for small IoT devices

Powermat PMT 100 is a wireless charging solution that allows small IoT devices to be charged more securely at a faster speed. This hybrid solution utilises both inductive wireless charging and resonance, thus producing more power, more freedom of positioning, longer ranges between wireless power

transmitters and receivers, and lower implementation costs. The module does not require any external receiver, microcontroller, and needs minimal



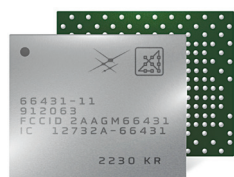
peripheral components, thus reducing design complexity and cost while

also reducing the size of the product. Powermat PMT 100 enables higher performance levels and a seamless charging user experience at a price point that supports scale. It is suitable for use in various applications, including small medical, lifestyle, and smart home Internet of Things (IoT) devices.

Powermat
<https://powermat.com>

World's smallest LTE-M/NB-IoT connectivity platform

Skyworks and Sequans have jointly developed SKY66431, the world's smallest LTE-M/NB-IoT connectivity platform for IoT communication systems. The SKY66431 is a highly



integrated multi-band, multi-chip SiP supporting 5G massive IoT platforms, offering

high-performance connectivity with ultra-low power consumption. The SiP can reduce the design complexity while providing high-performance end products, reducing the cost of development and accelerating time-to-market. The IoT platform is suitable for use in utility meters, asset trackers, security and alert systems, and other battery-powered devices, such as wearable

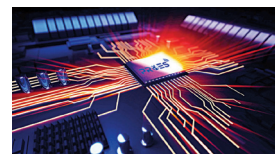
medical and fleet management.

Skyworks: <https://www.skyworksinc.com>

Sequans: <https://www.sequans.com>

RF booster co-processor for software or hardware implementation

PROES RF booster co-processor (RBcP) can improve reliability, communication distance, and power consumption



in any RF communication system. The RBcP enables designers to

modify the communication range without needing to change the bandwidth or the transmission power, therefore changing the communication range without changing any hardware or design. The longer transmission range lowers the number of repeaters, thus reducing the time and costs of deployment. PROES RBcP is available on software to be integrated in a software designed radios. For hardware implementation, the same functionality is also provided on FPGA.

PROES
<https://rfproes.com>

PMIC optimised for OLED IT devices

Magnachip's power management IC (PMIC) is designed for IT devices equipped with OLED screens. The PMIC



communicates with a system by using I2C protocol. Magnachip's

PMIC can control functions such as output voltage control, on/off block,